Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-6)

PHARMACEUTICAL BIOTECHNOLOGY-THEORY

Subject Code : BP605T

M.Code : 77990

Date of Examination : 20-05-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

   i. Role of DNA ligase

   ii. Applications of restriction endonucleases

   iii. Functions of Catalase and Proteases

   iv. ELISA

   v. Antitoxins

   vi. T Cell mediated immunity

   vii. Fed batch fermentation

   viii. Crystalloids

   ix. Plasmids

   x. Protein engineering.

SECTION-B

2. Describe in detail the working and applications of biosensors.

3. Write in detail the production of viral vaccines.

4. Write an explanatory note on factors affecting the design of media for fermentation. Highlight the role of agitation and aeration in fermentation .

SECTION-C

5. Emphasize the role of plasma substitutes with examples.

6. Write a note on hybridoma technology.

7. Describe the various methods of enzyme immobilization.

8. Describe the role of genetic engineering in medicine.

9. Describe the types of hypersensitivity reactions and their significance.

10. Describe the types of reactions mediated by microorganisms.

11. Write a note on the process of Transformation.

12. Describe the critical factors in the production of citric acid .

13. Illustrate the structure of Immunoglobulins.

LIBRARY (Mohali)

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.